

# NCDC Chatbot - Technical Architecture & Component Inventory

## Recommended Technology Stack

PDF Parsing: PyMuPDF  
Vector Database: Qdrant  
Metadata/User Database: PostgreSQL  
Chunking: Structure-Aware Parent Child Chunking  
Embeddings: BAAI/bge-m3  
Retrieval: Hybrid Search (BM25 + Dense Retrieval)  
Reranking: BAAI/bge-reranker-v2-m3  
LLM: GPT-4o-mini (MVP)  
Framework: LangGraph  
Cache: Redis  
Deployment: Docker + Linux VPS

## Frontend Modules

Chat Window, Chat Input Box, New Chat, Conversation History, Suggested Questions, Copy Response, Feedback Buttons, Citation Panel, Source Links, Follow-up Questions, Language Selector, Admin Dashboard, Analytics Dashboard.

## Authentication

admins table: id, name, email, password\_hash, role, created\_at, updated\_at

## Document Management

documents table: id, document\_name, file\_path, category, version, status, uploaded\_by, created\_at, updated\_at

## Document Processing

PDF Loader, Text Extractor, Metadata Extractor, Chunk Generator, Embedding Generator, Indexing Engine

## Chunk Storage

document\_chunks: chunk\_id, document\_id, page\_number, section\_name, chunk\_text, embedding\_id

## Qdrant Collection

Collection: ncdc\_documents Metadata: document\_name, page\_number, section\_name, category, version, chunk\_id

## Retrieval Pipeline

User Query → Query Processor → Hybrid Search (BM25 + Vector Search) → Result Merger → BGE Reranker → Context Builder → GPT-4o-mini → Citation Engine → Response

## Citation Engine

Citation Builder, Page Mapper, Section Mapper, Hyperlink Generator

## LangGraph Nodes

Start Node, Question Classifier, Query Rewriter, Hybrid Search Node, Reranker Node, Context Builder Node, LLM Node, Citation Node, Response Formatter Node, End Node

## Redis Usage

chat\_session, retrieval\_cache, embedding\_cache, prompt\_cache, rate\_limit\_cache

## Analytics Tables

chat\_logs, user\_feedback, analytics\_events, audit\_logs, chat\_sessions

## API Endpoints

POST /login  
POST /logout  
POST /upload-document  
GET /documents  
PUT /document  
DELETE /document  
POST /chat  
POST /feedback  
GET /chat-history  
GET /analytics  
GET /feedback-summary

## Production Architecture

Frontend → FastAPI → LangGraph → Hybrid Retrieval → Qdrant + BM25 → BGE Reranker → GPT-4o-mini → Citation Engine → Final Response